

“Rewards and Consumption in the Credit Card Market” by Tianyu Han

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Credit Card Context

Many papers documenting behavioral mistakes in the credit card market

- Focus on costs of mistakenly incurring interest or late fees

(Beshears, Choi, Laibson, Madrian 18 Handbook Chapter)

- Increasing recent focus on rewards

(e.g., Mukharlyamov & Sarin, 24; Wang 24; Huynh, Nicholls and Shcherbakov 22; Agarwal, Presbitero, Silva, and Wix 23; Agarwal, Ang, Wang, and Zhang 24)



Data and Methods

Administrative data from large Chinese bank who offers credit cards.

RDD for platinum reward card eligibility to estimate effects of rewards on spending.

(“Fuzzy RDD is IV”, Angrist and Pischke, 2009)

Use linked survey of 4,565 credit card users. Response rate?

(see Stantcheva 22 Ann. Review of Economics; Bergman, Chinco, Hartzmark, & Sussman 20)

Structural model evaluating implications for market structure & welfare.

Han's Hypothesis

1. Rewards → Spend More on Reward *and on Non-Reward* Categories

Reward spending \$64 (30%?), non-reward spending \$54 (5%?)

2. Why ↑ **Non-Reward Spending?** 'Complementary Ignorance'
(naïve consumers under-estimate how rewards change non-reward spending)

e.g., airline rewards → book flight.

But don't consider also spend on non-reward category (e.g. car rental)

3. Implications?

(i) Incentive banks to offer more generous rewards

(ii) Cross-subsidizes less to more sophisticated

(iii) **2.5% consumption cost per month**

Big Picture

- **Impressive data collection**
- **Convincing RDD design to show that rewards “work”**
- **Interesting new theory**

Three Discussion Points:

- 1. Rewards**
- 2. Mistakes**
- 3. Implications**

Linking Reward Effects to Literature

Good job of linking to Bursztyn, Ferman, Fiorin, Kanz, & Rao (17 QJE)

- platinum cards are status goods & ↑ in-person spending

- Han's paper shows effect on online transactions.



Gelman & Roussanov (24 RFS):

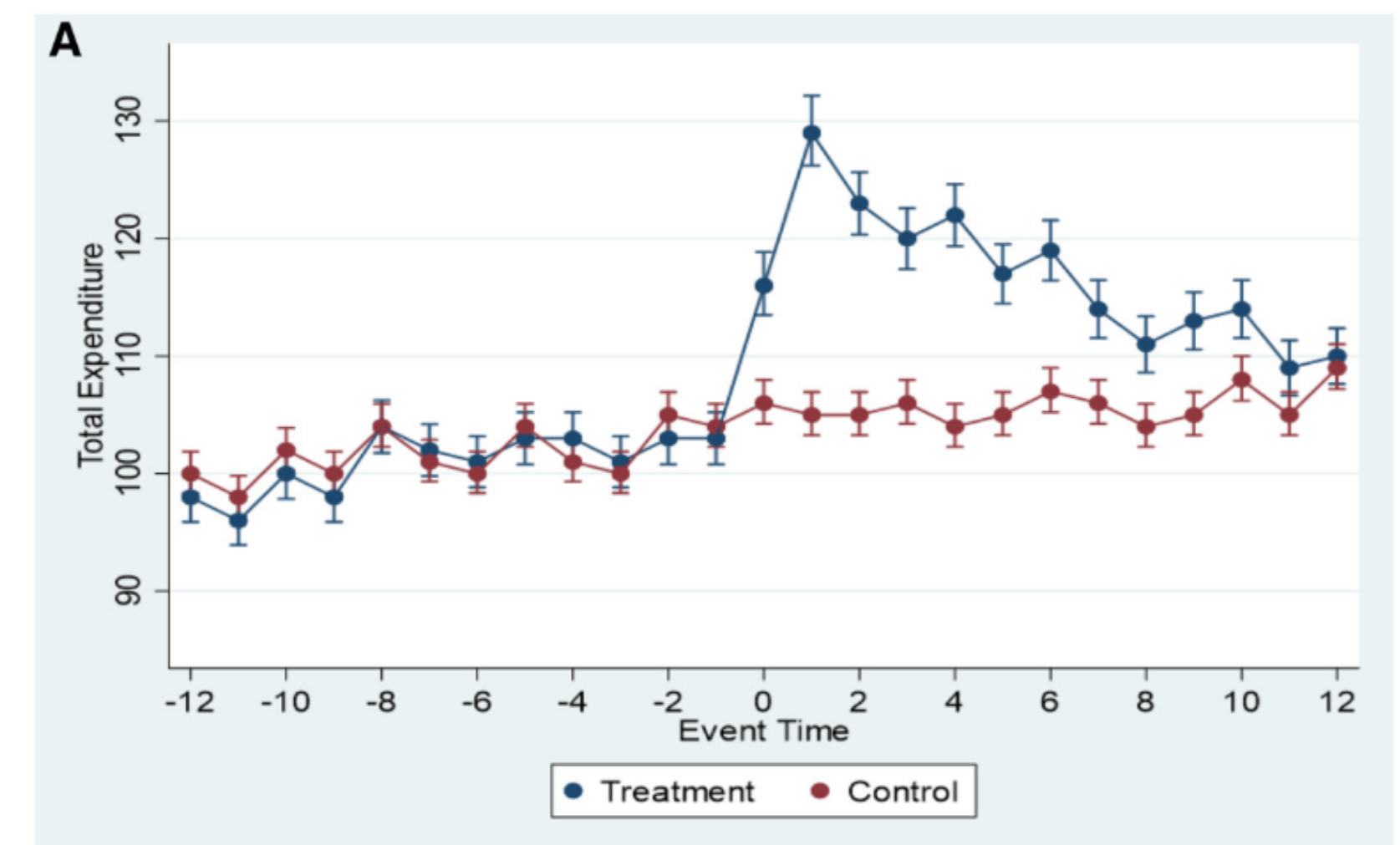
New credit card -> ↑ credit card spending

- What is the marginal value of rewards?

Rewards ↑ spending is consistent with US evidence

(Agarwal, Presbitero, Silva, & Wix 23; Agarwal, Ang, Wang, & Zhang 24)

- Decompose non-reward effects into credit card & non-credit card?



#1: Platinum vs. Gold Rewards

Table 1. Example of Credit Card Rewards

	Gold	Platinum
5% off JD.com purchases	Y	Y
50% Starbucks/KFC	Y	Y
5% off gas/groceries	Y	Y
\$10 off movie tickets	Y	Y
Cashback on international flights		Y
Foreign airport pickup		Y
Travel insurance		Y
Hotel free buffet		Y
Travel medical insurance		Y

- Rewards (slightly?) reduce price of travel
- Are platinum rewards only for travel?
- Which rewards are % spending?
- Which are salient?
- Which MCCs drive “non-reward” results?
- Beyond scope of this paper...
how do consumers value different rewards?

#2 Mistakes

What I expected:

1. Survey expectations of future spending
2. Exogenous variation in platinum cards
3. Compare ex-ante expectations vs. realized spending

What this paper does:

1. Study causal effects on historical spending
2. Survey understanding of realized spending in last 6 months.

#2 Mistakes

“Credit card rewards can lure naive consumers into booking reward-earning goods and services while overlooking the complementary purchases that will be necessary in the future.”

Consumers less aware of total spending is *different* to consumers not planning their non-rewards expenditure & overconsuming.

Spending in last 6 months is a mix of consumers (average 282 days with current card):
(i) 6+ months platinum (ii) <6 months platinum (iii) Never platinum.

Is there learning? Story most convincing if persistent errors among experienced users.

→ Heterogeneity has noisy nulls for longer-holding & older users.

Total spend under-reporting: \$82 (s.e. \$51) & \$60 (s.e. \$44)

#3 Implications

Model shows welfare loss. What about counterfactual policies?

- Interchange caps of different levels (5.25% baseline)
 - Simplify reward structure (e.g., can only be X% of any spending)
- (Iberti, Kulkarni, & Truffa, 23; US regulations do *not* standardize reward disclosures)

More generally, complexity in credit card rewards appears productive area to explore:

- How difficult to choose optimal credit card to maximize benefits?
- For credit cards held, how do cardholders allocate spending to maximize benefits?

Many reward categories. Different units: points, miles, cash. Annual fees. Bundled benefits.

(Agarwal, Chomsisengphet, Liu 10 JMCB; Agarwal, Chomsisengphet, Liu, Souleles 15 RCFS; Stango & Zinman, 16 RFS; Ru & Schoar, 23; CFPB 24)

Thanks!

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